

Array Decomposition

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TBD

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Note: This is a working manuscript which will be expanded/updated frequently. All suggestions for improvement are welcome. All qmd, tex, html, pdf, R, and C files are in the public domain. Attribution will be appreciated, but is not required. The files can be found at <https://github.com/deleeuw/arrays>

1 Introduction

2 Decomposition

2.1 Vectors

Suppose x is an element of n -dimensional Euclidean space, with the usual inner product $x'y$ and the squared norm $\|x\|^2 := \sqrt{x'x}$. We can decompose x into two orthogonal parts, with the first part a vector with all elements equal to the mean and the second part a vector with the deviations from the mean. We use a $\bullet$ replacing a subscript to indicate averaging over that subscript. The elementwise decomposition is

$$x_i = x_\bullet + (x_i - x_\bullet). \quad (1)$$

Equation (1) can be written as

$$x = Px + (I - P)x, \quad (2)$$

where P projects on the subspace of all constant vectors, i.e. $P := n^{-1}ee'$ with e a vector with all elements equal to one. Now $P^2 = P$ and thus $P(I - P) = 0$, which shows that Px and $(I - P)x$ are orthogonal. Also, as Pythagoras used to say,

$$\|x\|^2 = \|Px\|^2 + \|I - Px\|^2. \quad (3)$$

Note that equation (2) is actually correct for any square matrix P , but the two components are orthogonal if and only if P is a projector. Thus (3) is also only true if P is a projector.

We can generalize equation (2) to mutually orthogonal subspaces $S_1, \dots, S_m$, with corresponding projectors $P_1, \dots, P_m$. If $P_1 + \dots + P_m = I$ we can have the orthogonal decomposition

$$x = P_1x + P_2x + \dots + P_mx, \quad (4)$$

and if $P_1 + \dots + P_m \lesssim I$ in the Loewner sense¹ then

$$x = P_1x + P_2x + \dots + P_mx + (I - P_1 - \dots - P_m)x \quad (5)$$

Note that the “residual” $I - P_1 - \dots - P_m$ is orthogonal to all $P_1, \dots, P_m$.

Suppose U is an $n \times n$ matrix with $U'U = nI$ and with first column equal to e . Define $\alpha := n^{-1}U'x$ and $P_j := n^{-1}u_ju_j'$. The P_j are projectors orthogonal to each other. Then

$$U\alpha = \sum_{j=1}^n P_jx = \left(\frac{1}{n}ee'\right)x + \left(I - \frac{1}{n}ee'\right)x = x_\bullet e + (x - x_\bullet). \quad (6)$$

¹The eigenvalues of $P_1 + \dots + P_m$ are less than or equal to one.

Although α depends on the choice of U the product $U\alpha$ is the same for all U .

We can also derive the decomposition from a least squares projection perspective. Suppose we want to minimize $\|x - a\|^2$ over all a of the form $a = \alpha e$. The solution for α is, of course, $Px = x_{\bullet}$. And we may want to minimize $\|x - a\|^2$ over all a satisfying $e'a = 0$, which gives the solution $(I - P)x = x - x_{\bullet}$.

The projection formulation suggests some obvious further generalizations. We can replace the one-dimensional subspace of all constant vectors by an arbitrary subspace S , and we can replace the inner product $x'y$ by an arbitrary inner product $x' Ay$, with A positive semi-definite. If H is an A -orthonormal basis for the subspace S , i.e. if $H'AH = I$, then $Px = HH'Ax$ or $(x - Px)'Az = 0$ for all $z \in S$.

In the next sections we analyse the same approaches we have used for vectors to decompositions of linear spaces of matrices and arrays. This is actually not as much of a generalization as it appears to be, because matrices and arrays are elements of inner product spaces, just like vectors are. What we will introduce are different notations and in most cases different inner products.

2.2 Matrices

Suppose X is an $n \times m$ matrix, an element of the space $\mathbb{R}^{n \times m}$ with inner product $\langle X, Y \rangle := \text{tr } X'Y$ and squared norm $\|X\|^2 = \text{tr } X'X$. The elementwise decomposition we are interested in is

$$x_{ij} = x_{\bullet\bullet} + (x_{i\bullet} - x_{\bullet\bullet}) + (x_{\bullet j} - x_{\bullet\bullet}) + (x_{ij} - x_{i\bullet} - x_{\bullet j} + x_{\bullet\bullet}). \quad (7)$$

Using ANOVA terminology we have the *overall effect*, the two *main effects* for rows and columns, and the *first order interaction* of the rows and columns.

We define the projectors

$$P_r := n^{-1}ee', \quad (8a)$$

$$P_c := m^{-1}ee', \quad (8b)$$

and with $Q_r := I - P_r$ and $Q_c := I - P_c$. Equation (7) becomes

$$X = P_rXP_c + P_rXQ_c + Q_rXP_c + Q_rXQ_c. \quad (9)$$

Thus again we have four projectors, orthogonal to each other, adding up to the identity.

In de Leeuw (2026) we generalize the results in this section to the inner product $\langle X, Y \rangle := \text{tr } RXY'C$, with R and C positive semi-definite. Decomposition (9) remains correct, with $P_r = ee'R/e'Re$ and $P_c = Cee'/e'Ce$. Note that both projectors are asymmetric. Equation (7) also remains valid, if we interpret the bullitt subscripts as weighted means.

To see more clearly how the vector case and the matrix case are different we use the $\text{vec}()$ operator to map $\mathbb{R}^{n \times m}$ into $\mathbb{R}^{nm}$. Using the Kronecker product $\otimes$ of matrices we have the identity

$$\text{tr } RXY' = x'(C \otimes R)y, \quad (10)$$

with $x := \text{vec}(X)$ and $y = \text{vec}(Y)$. Thus our matrix results can be interpreted as a special case of the vector results in $\mathbb{R}^{nm}$, with inner product $x'(C \otimes R)y$.

Define U and V to be square matrices with first column equal to one, $U'U = nI$ and $V'V = mI$. Then define

$$\alpha_{ij} = \frac{1}{nm} u_i' X v_j.$$

Now

$$X = \sum_{i=1}^n \sum_{j=1}^m \alpha_{ij} u_i v_j'$$

$$\|X\|^2 = \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^n \sum_{l=1}^m a_{ij} a_{kl} \text{tr } v_j u_i' u_k v_l' = \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^n \sum_{l=1}^m \delta^{ik} \delta^{jl} a_{ij} a_{kl} = \|A\|^2$$

Term with $i = 1$ and $j = 1$.

$$a_{11} e e' = x_{\bullet\bullet} e e'$$

Terms with $i = 1$ and $j \neq 1$.

$$\frac{1}{nm} \sum_{j=2}^m \alpha_{1j} e v_j' = \frac{1}{nm} e \sum_{j=2}^m (e' X v_j) v_j' = n^{-1} e e' X J_m = e(x_{\bullet_j} - x_{\bullet\bullet})'$$

$$\frac{1}{nm} \sum_{i=1}^n \sum_{j=2}^m (u_i' X v_j) u_i v_j' = J_n X J_m = (x_{ij} - x_{i\bullet} e' - e x_{j\bullet} + x_{\bullet\bullet})$$

2.3 Arrays

If X is an $n \times m \times \ell$ array then the elementwise decomposition we are interested in is

$$x_{ijk} = x_{\bullet\bullet\bullet} + (x_{i\bullet\bullet} - x_{\bullet\bullet\bullet}) + (x_{\bullet j\bullet} - x_{\bullet\bullet\bullet}) + (x_{\bullet\bullet k} - x_{\bullet\bullet\bullet}) \quad (11)$$

$$+ (x_{ij\bullet} - x_{i\bullet\bullet} - x_{\bullet j\bullet} + x_{\bullet\bullet\bullet}) \quad (12)$$

$$+ (x_{i\bullet k} - x_{i\bullet\bullet} - x_{\bullet\bullet k} + x_{\bullet\bullet\bullet}) \quad (13)$$

$$+ (x_{\bullet j k} - x_{\bullet j\bullet} - x_{\bullet\bullet k} + x_{\bullet\bullet\bullet}) \quad (14)$$

$$+ (x_{ijk} - x_{ij\bullet} - x_{i\bullet k} - x_{\bullet j k} + x_{i\bullet\bullet} + x_{\bullet j\bullet} + x_{\bullet\bullet k} - x_{\bullet\bullet\bullet}) \quad (15)$$

3 Interactions

4 Orthogonal Functions

This section is inspired by an anonymous and unpublished manuscript that I read around 1980 with a title that might have been “Lois de Probabilité sur un Ensemble Produit” and perhaps the words “Décomposition” and “Interaction” were also in the title or subtitle. It was either by Jean-Paul Benzécri or by one of his students or co-workers. I did have a copy and I lost it, and Google can’t find me a new one. The published papers by (?) and (?) discuss the same materials, but in less detail.

Suppose M_n is the set of matrices X of order n with

- first column all elements equal to one, and
- $X'X = nI$.

Suppose the array A is $n \times m \times \ell$. Choose $X \in M_n$, $Y \in M_m$, and $Z \in M_\ell$. Define

$$\alpha_{ijk} = \sum_{i'=1}^n \sum_{j'=1}^m \sum_{k'=1}^l a_{i'j'k'} x_{i'i} y_{j'j} z_{k'k}$$

$$\alpha_{ij1} = \sum_{i'=1}^n \sum_{j'=1}^m a_{i'j' \bullet} x_{i'i} y_{j'j}$$

References

de Leeuw, Jan. 2026. “Matrix Decomposition with Kronecker Weights.” <https://jansweb.netlify.app/publication/deleeuw-e-26-e/deleeuw-e-26-e.pdf>.